

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Full-stack engineer with production experience in Python backend systems and React/Next.js frontends. Built and deployed REST APIs, Auth0 authentication, S3 file pipelines, and Docker-containerized services across multiple production systems. Python work spans FastAPI semantic search APIs, AI workflow automation at Mphasis, and LLM evaluation harnesses. IEEE-published research in structured knowledge extraction: T5 transformer pipeline converting 1,500+ PDFs into structured mind maps and flowcharts -- directly relevant to turning fragmented documents into reusable knowledge assets. MS CS at NC State (May 2026). Portfolio: kulkarniaditya.com

CORE COMPETENCIES

Python + FastAPI

React + Next.js

REST APIs

Auth0 + OAuth/RBAC

Docker + Linux

AWS + S3

Automated Testing

CI/CD + GitHub Actions

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Built 10+ production backend features in TypeScript and Node.js: REST API endpoints, Auth0 OAuth/RBAC authentication, S3 file storage pipelines, email parsing via Microsoft Graph API, and real-time event delivery -- **394 commits, 7 production deployments**, full SDLC ownership
- Implemented Temporal.io workflow orchestration for reliable at-least-once execution of AI agent pipelines across 3 LLM providers (Claude, GPT-4o, Gemini); managed Docker containerization, PostgreSQL + Redis data layers, Sentry observability, and GitHub Actions CI/CD

NC State University Apr 2025 - Present

Graduate Research Assistant

- Deployed a cloud-native genomic data platform on GCP Cloud Run (containerized MERN stack), enabling interactive exploration of 500GB scientific datasets for 100+ researchers; applied Python-based hyperparameter tuning and time-series analysis improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven Python workflows including OCR, NLP, and ML automation for health claim processing, improving accuracy by 25%; automated AWS and GCP deployments with Terraform and Grafana monitoring dashboards

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built Python-based distributed data pipelines (Hadoop + Spark batch, Kafka streaming) reducing ticket-processing latency by 20%; led SDLC for a data-driven backend system reducing processing time by 15%

PROJECTS

VectorLift -- Semantic Search and Ranking Engine Python + FastAPI + Docker

Built a production Python semantic search system: BM25 baseline (Elasticsearch), dense bi-encoder retrieval (sentence-transformers fine-tuned on MS MARCO), and cross-encoder re-ranker (MiniLM). Served via FastAPI REST API with per-stage latency tracking; Kafka-driven indexing pipeline for corpus ingestion into Elasticsearch and Qdrant vector store; Prometheus + Grafana observability. Full Docker Compose orchestration. Evaluated across 6 configurations with NDCG@10, MRR@10, MAP.

Stack: Python, FastAPI, PyTorch, Elasticsearch, Kafka, Qdrant, PostgreSQL, Redis, Prometheus, Grafana, Docker Compose

PulseRank -- Real-Time Leaderboard and Activity Feed Next.js + NestJS + REST APIs

Full-stack platform with a Next.js frontend and NestJS REST API backend: MySQL transactional writes with idempotency-key deduplication and outbox pattern, MongoDB feed projections with TTL expiry, Redis sorted sets for leaderboard reads, Server-Sent Events for live updates. Testcontainers integration tests verifying multi-store consistency; k6 load benchmarking; OpenTelemetry + Prometheus observability; Docker + GitHub Actions CI/CD.

Stack: TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, GitHub Actions

Mapify -- ML and NLP Document Structure Extraction IEEE Published + Knowledge Graph

IEEE-published research: built a T5 transformer pipeline that converts 1,500+ PDFs and images into structured mind maps and flowcharts -- extracting relationships, entities, and hierarchies from unstructured text. Used NLTK for tokenization, NetworkX for graph construction, reducing academic document review time by 60%. Presented to 500+ attendees. Directly applicable to BCC's mission of turning fragmented documents into structured, reusable knowledge assets.

Stack: Python, T5 Transformer, NLTK, NetworkX, HuggingFace, PDF processing

Glance AI -- Privacy-First RAG System Python + Knowledge Retrieval

Built a compliance-ready RAG system for fintech and healthcare environments: per-tenant data isolation at the vector retrieval layer preventing context leakage, deterministic output architecture ensuring factual correctness for regulated data. Designed for explainable AI requirements in sensitive domains -- aligns directly with BCC's focus on trustworthy AI and structured knowledge.

Stack: Python, LLMs, Vector Databases, RAG, per-tenant isolation

EDUCATION

North Carolina State University -- Master of Science, Computer Science Aug 2024 - May 2026

Courses: AI, Operating Systems, Data Science, Algorithms, Software Engineering, Object Oriented Development

Mumbai University (DJSCE) -- Bachelor of Technology, Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 | IEEE-published NLP research (Mapify); Top 10 of 100+ teams HackNC24

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C Frontend: React, Next.js, Tailwind CSS

Backend: FastAPI, Node.js, NestJS, Express, REST APIs, Temporal.io, Kafka Auth: Auth0, OAuth 2.0, JWT, RBAC, session management

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, SQLite

Cloud + Infra: AWS (S3, EC2), GCP (Cloud Run), Docker, Kubernetes, Terraform, GitHub Actions

Testing: Testcontainers, integration tests, unit tests, k6 load testing

AI/ML: PyTorch, HuggingFace, sentence-transformers, LangChain, RAG, FAISS, Qdrant, OpenAI/Anthropic/Gemini APIs